

Make the entire UI design and every component mobile responsive.
Use tailwind CSS

Navigation type

Collapsible sidebar — the hamburger button collapses/expands it. On mobile (narrow screens) it becomes a slide-over drawer with a backdrop overlay so the main content always has full width.

Dashboard page

The dashboard now flows as: metrics → field selector → charts → density map

Metrics — The grid is now 4 columns, collapsing to 2×2 on smaller screens. Device status is added to the metrics row, showing each device ID with a connectivity dot and an idle/working/offline tag

Field selector bar — dropdown at the top lets you switch between fields. "Manage fields" opens a modal listing all fields with delete options. "+ Add field" opens the creation modal.

Add field modal — inputs for field name, width × height in metres, partition type (row or column), and partition count. A live preview line summarises the resulting layout.

Charts-weed trend line chart (left) paired with a species donut pie chart (right), and below that **Density map**-full 2×2 ft field grid density map with green/orange/red heat encoding.



Dashboard



Historical data



Device control

Total weeds detected
3,284

Total runs
47

Active devices
2 / 3

Device status

DEV-001

Working

DEV-002

Idle

DEV-003

Offline

FIELD



North Field — 40×30 m



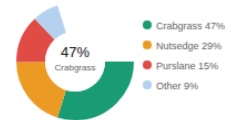
Manage fields

+ Add field

WEEDS DETECTED PER RUN

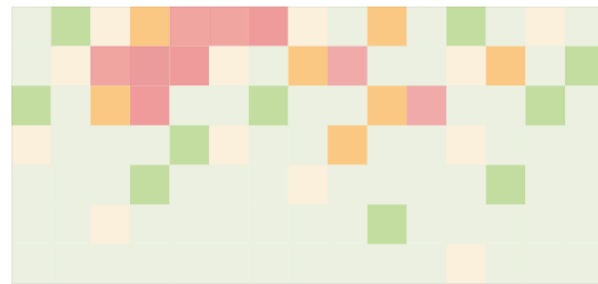


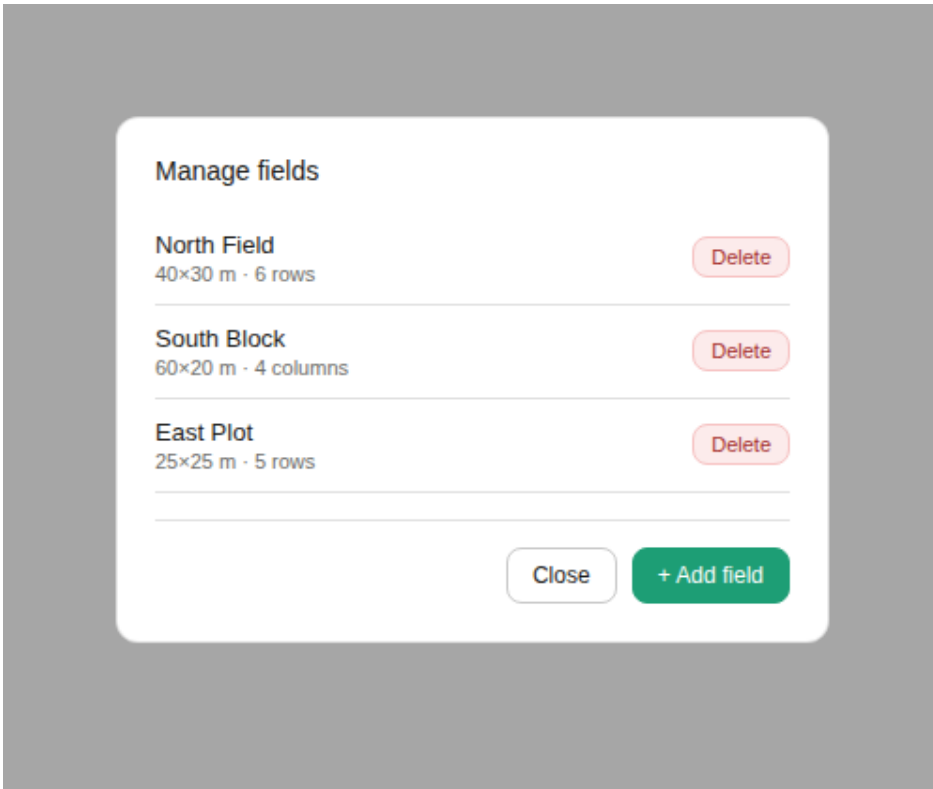
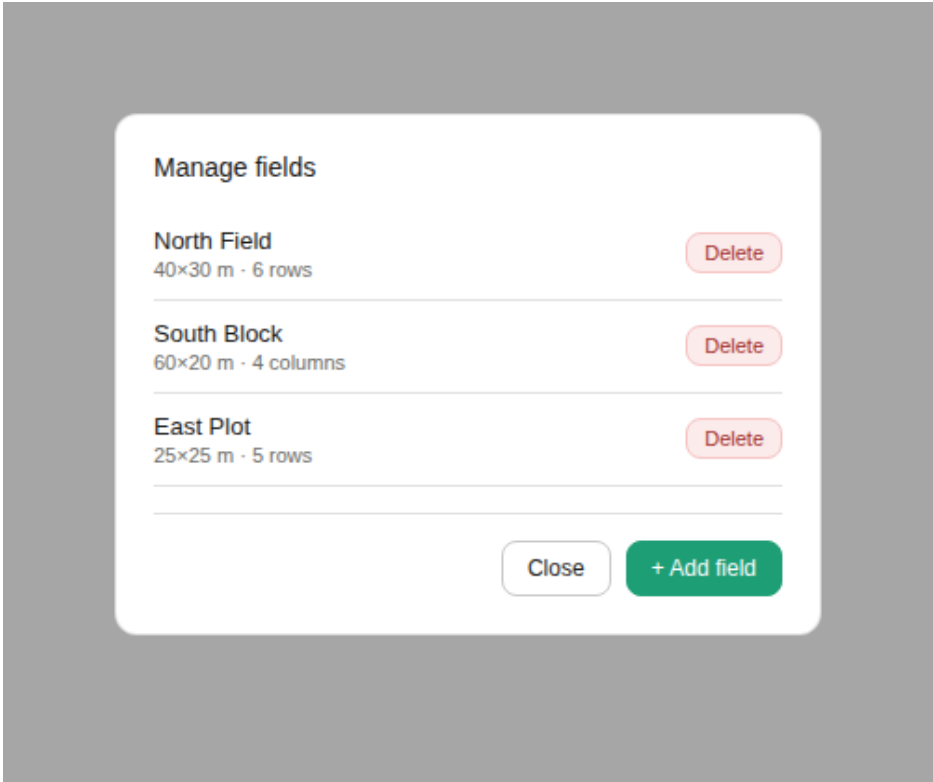
WEED SPECIES BREAKDOWN



WEED DENSITY MAP · NORTH FIELD · 2×2 FT GRID

Low Medium High





Analytics page :

Analytics page — a horizontal filter bar at the top with device ID, field, and month dropdowns that can be used individually or combined, with active filter chips showing what's applied. Below is a paginated session table with a rows-per-page selector.

View Session run page (opens on row click, simulating a new page navigation) — four sections stacked:

- Run header card with title, date, field, duration and weed count badge
- Field weed distribution grid + species breakdown donut with bar chart, side by side
- Device analytics table — device ID, total weeds detected, total photos taken, total run time
- Detection logs table — grid position, original photo thumbnail, annotated photo thumbnail (with red bounding box indicators), species tag, and coordinates — fully paginated



Dashboard

Analytics

Device control

Device ID

Field

Month

All devices

All fields

All months

Apply

Reset

Active filters:

DEV-001

April 2025

SESSION LOG

Showing 1-8 of 23 results

Run	Device ID	Field	Date & time	Weeds	Duration	
#47	DEV-001	North Field	Apr 5, 2025 08:32	124	58m	View
#46	DEV-001	North Field	Apr 3, 2025 14:10	87	42m	View
#45	DEV-002	South Block	Apr 2, 2025 09:05	32	31m	View
#44	DEV-001	North Field	Mar 30, 2025 11:22	61	38m	View
#43	DEV-002	East Plot	Mar 28, 2025 07:44	18	22m	View
#42	DEV-001	North Field	Mar 26, 2025 10:15	73	44m	View
#41	DEV-003	South Block	Mar 24, 2025 15:50	49	35m	View
#40	DEV-002	East Plot	Mar 22, 2025 08:00	95	51m	View

Rows per page:

8

<

1

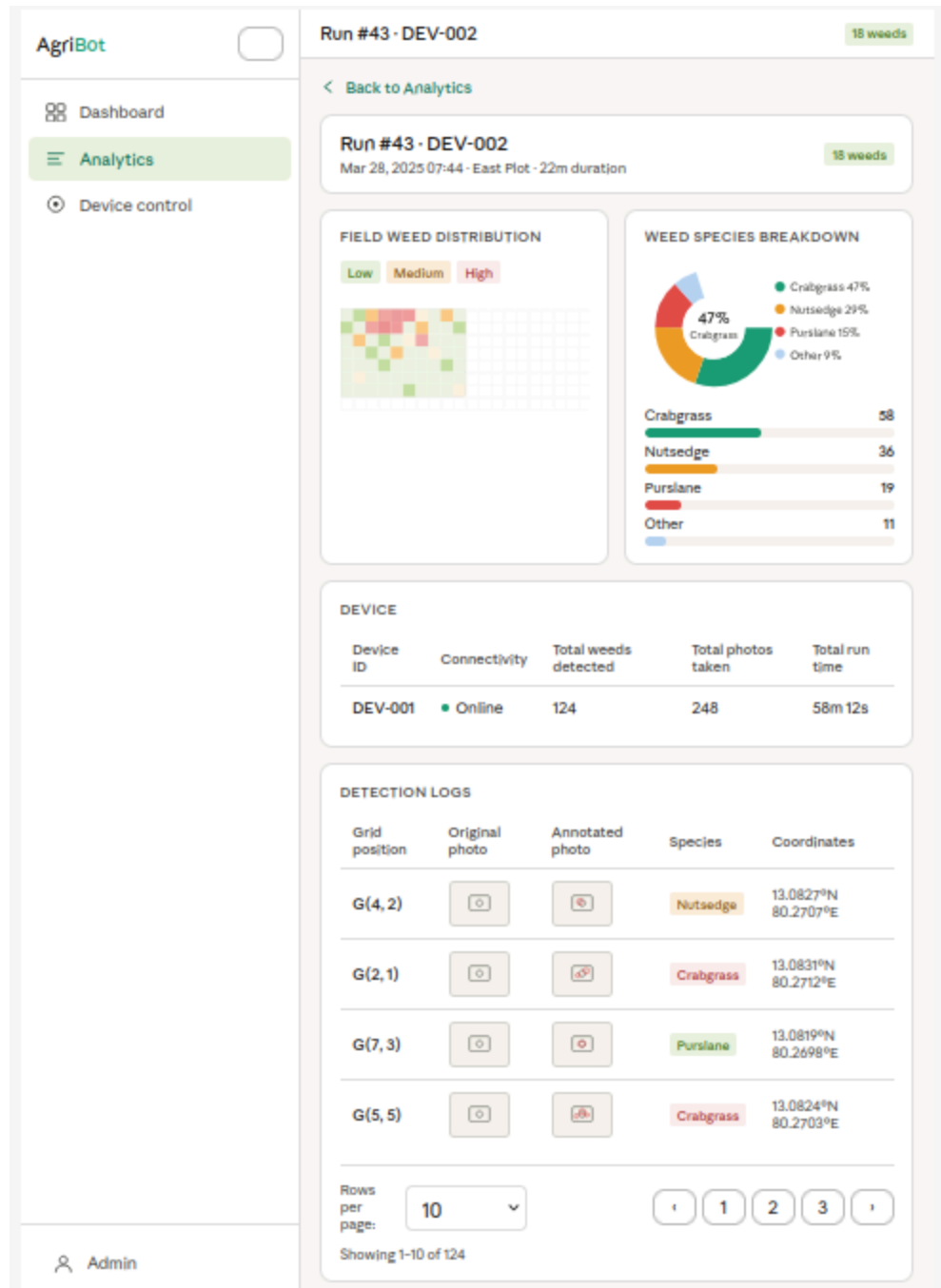
2

3

>

Page 1 of 3

Admin



Device Control page :

Select device & field— choose a device ID and field from the two dropdowns. Once selected, a status strip appears showing connectivity and idle/working/offline status. DEV-003 (offline) will disable the start button.

The "Controlling: DEV-00X · Last run..." note appears below.

Weed detection — toggle between Grid position and Select route via radio-style buttons. The start button only activates once a device and field are both selected. Click **Start weed detection** to see: the panel switches to a live field map with an animated device position dot, cells scanned and weeds found counters update in real time, and manual control gets disabled with a warning notice. Click **Stop weed detection** to return to the idle view — the last run note updates to "stopped by user at..."

Manual control — a directional D-pad for quick movement, with distance (metres) and angle (degrees) inputs below. Four action buttons map to forward, backward, turn left, and turn right.

Route designer — command + value dropdowns to build a step list. Click "+ Add" to append instructions, "Remove" to delete individual steps (numbers auto-renumber), then name and save the route. Rename start weed detection command to detect weed and it shows distance(metre)

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Dashboard

Analytics

Device control

Device control

3 devices

SELECT DEVICE & FIELD

Device ID

DEV-001

Field

North Field — 40×30 m

DEV-001

Connectivity: Online

Idle

Field: North Field

Controlling: DEV-001

No runs yet

WEED DETECTION

Grid position

Select route

Grid X

e.g. 4

Grid Y

e.g. 2

Forward distance (metres)

e.g. 5.0

Start weed detection

MANUAL CONTROL

↑

←

→

↓

Distance (m)

0.5

Turn angle (°)

90

Move fwd

Move bwd

Turn left

Turn right

DESIGN ROUTE

Command

Move forward

Distance (m)

e.g. 2.5

+ Add

1

Move forward - 3 m

Remove

2

Turn right - 90°

Remove

3

Move forward - 3 m

Remove

4

Detect weed - 2 m

Remove

Route name

e.g. Route A

Save route

Clear all

Admin

8

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Dashboard

Analytics

Device control

Device control

3 devi ...

SELECT DEVICE & FIELD

Device ID

DEV-001

Field

North Field — 40×30 m

DEV-001

Connectivity: Online

Idle

Field: North Field

Controlling: DEV-001

No runs yet

WEED DETECTION

Grid position

Select route

Select route

Route A — 6 steps

Route A: Forward 3m → Turn right 90° → Forward 3m → Detect weed 2m → ×3

Start weed detection

MANUAL CONTROL

↑

←

⊞

→

↓

Distance (m)

0.5

Turn angle (°)

90

Move fwd

Move bwd

Turn left

Turn right

DESIGN ROUTE

Command

Move forward

Distance (m)

e.g. 2.5

+ Add

1 Move forward - 3 m

Remove

2 Turn right - 90°

Remove

3 Move forward - 3 m

Remove

4 Detect weed - 2 m

Remove

Route name

e.g. Route A

Save route

Clear all

Admin

Settings Page :

Step 1 — Select device: choose DEV-001 or DEV-002 to see the device status strip (online/idle). Selecting DEV-003 (offline) keeps Fetch settings disabled.

Step 2 — Fetch settings: click Fetch settings to see a spinner, then the configuration table appears with all 6 values: `device_id`, `server_url`, `serial_port`, `serial_baud_rate`, `camera_index`, `confidence_threshold`. Each row has an **Edit** button that switches the value to an inline input with Save/Cancel.

Step 3 — Send settings: the Send settings button enables only when the device is connected and idle. Click it to see the spinner progress, then a green success banner ("Settings sent successfully — device updated.") or a red failure banner appears below.

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Dashboard

Analytics

Device control

Settings

Admin

Settings

DEVICE

Select device

— Choose device —

Fetch settings

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Dashboard

Analytics

Device control

Settings

Admin

Settings

DEVICE

Select device

DEV-003 (offline)

Fetch settings

DEV-003

Connectivity: Offline Offline

AgriBot

Dashboard

Analytics

Device control

Settings

Admin

Settings

DEVICE

Select device

DEV-001

Fetch settings

DEV-001

Connectivity: Online Idle

AgriBot

Dashboard

Analytics

Device control

Settings

Admin

Settings

DEVICE

Select device

DEV-001

Fetch settings

DEV-001

Connectivity: Online Idle

DEVICE CONFIGURATION

Click Edit to modify a value

device_id	DEV-001	Edit
server_url	http://192.168.1.10:5000	Edit
serial_port	/dev/ttyUSB0	Edit
serial_baud_rate	115200	Edit
camera_index	0	Edit
confidence_threshold	0.75	Edit

Device is connected and idle — ready to receive settings.

Send settings

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Dashboard

Analytics

Device control

Settings

Admin

Settings

DEVICE

Select device

DEV-001

Fetch settings

DEV-001

Connectivity: Online Idle

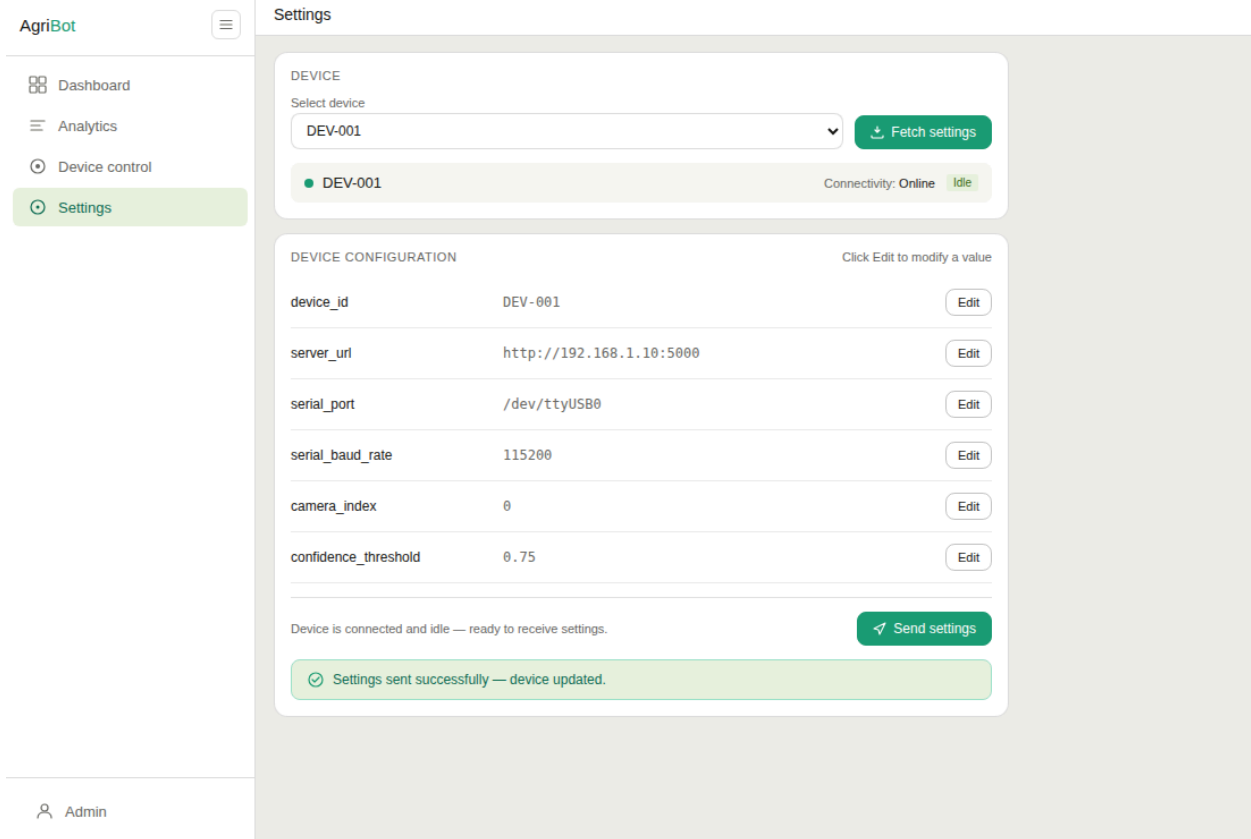
DEVICE CONFIGURATION

Click Edit to modify a value

device_id	DEV-001	Edit
server_url	http://192.168.1.10:5000	Edit
serial_port	/dev/ttyUSB0	Edit
serial_baud_rate	115200	Edit
camera_index	0	Edit
confidence_threshold	0.75	Edit

Sending settings to device...

Sending...



Settings page :

Filter bar — type in the device name search box to filter the table in real time, or pick a created date to narrow results. The "Showing X of Y" label updates accordingly. Click Clear to reset.

each filter now has its own inline × button that appears only when that field has a value. Clicking × on the device name clears only the name search, and clicking × on the created date clears only the date — each independently re-runs the filter without touching the other.

Device table — shows device name, connectivity dot + online/offline, created date, server URL, serial port, baud rate, and a Delete button per row.

Delete flow — clicking Delete opens a confirmation modal naming the specific device. Confirming removes it from the table immediately.

Create new device — click "+ New device" to open the creation modal. Key behaviours:

- **Device name check:** as you type, after a short debounce it checks against existing names and shows a live "✓ Available" or "× Already exists" indicator
- **All 6 config fields** are mandatory (`device_id`, `server_url`, `serial_port`, `serial_baud_rate`, `camera_index`, `confidence_threshold`) — each shows a red border and error hint if left empty
- **Create device button** stays disabled until the name is confirmed available AND all config fields are filled
- Newly created devices appear in the table immediately with today's date and offline status (until the physical device connects)

The page also has a "Device manager" entry added to the sidebar nav between Device control and Settings.

Here are all 5 screenshots covering the complete device manager flow:

1. **Device manager default** — full table with all 3 devices, empty filters, New device button
2. **Filter with inline clear** — both filters active (DEV-001 + date), each showing its individual × button inside the input, table filtered to 1 result
3. **Create modal — empty** — initial state, all fields blank, Create device button disabled
4. **Create modal — name taken** — "DEV-001" entered, red error state with "× Already exists" inline, error hint below
5. **Create modal — ready to submit** — "DEV-004" entered with "✓"

Device manager

3 devices

Search device name

e.g. DEV-001

Created date

mm/dd/yyyy



+ New device

ALL DEVICES

Showing 3 of 3

Device name	Connectivity	Status	Created date	
DEV-001	● Online	Working	2025-01-10	Delete
DEV-002	● Online	Idle	2025-02-14	Delete
DEV-003	● Offline	Offline	2025-03-22	Delete

Device manager

3 devices

Search device name

DEV-001



Created date

01/10/2025



+ New device

ALL DEVICES

Showing 1 of 3

Device name	Connectivity	Status	Created date	
DEV-001	● Online	Working	2025-01-10	Delete

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Device manager

3 devices

Device manager

Create new device

All fields are mandatory. Device name must be unique.

Device name *

e.g. DEV-004

Enter a name

Enter a unique device name

CONFIGURATION

device_id *

e.g. DEV-004

Must match the device name

server_url *

http://192.168.1.10:5000

Flask server endpoint URL

serial_port *

/dev/ttyUSB0

serial_baud_rate *

115200

camera_index *

0

confidence_threshold *

0.75

Value between 0 and 1

Cancel

Create device

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Device manager

Device manager

Create new device

All fields are mandatory. Device name must be unique.

Device name *

DEV-001

× Already exists

This device name is already taken.

CONFIGURATION

device_id *

e.g. DEV-004

Must match the device name

server_url *

http://192.168.1.10:5000

serial_port *

/dev/ttyUSB0

serial_baud_rate *

115200

camera_index *

0

confidence_threshold *

0.75

Value between 0 and 1

Cancel

Create device

AgriBot

Device manager

Device manager

Create new device

All fields are mandatory. Device name must be unique.

Device name *
DEV-004 ✓ Available
Device name is available.

CONFIGURATION

device_id *
DEV-004
Must match the device name

server_url *
http://192.168.1.13:5000

serial_port *
/dev/ttyUSB3

serial_baud_rate *
115200

camera_index *
0

confidence_threshold *
0.75
Value between 0 and 1

Cancel Create device